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Status	Finished
Started	Thursday, 20 November 2025, 1:30 PM
Completed	Thursday, 20 November 2025, 1:55 PM
Duration	24 mins 43 secs
Marks	3.00/3.00
Grade	3.00 out of 3.00 (100%)

Information

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This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Correct

Mark 1.00 out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (3713, 613)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$

Your last answer was interpreted as follows:

2113

It might help to know that the following is a list of the first prime numbers:

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$M = 3398$

Find the unencrypted (plain text) message

$m =$

Your last answer was interpreted as follows:

2055

The following calculator can be used for computing exponents $\text{mod } (n)$:

Base:	
Exponent:	
Modulus:	
Result:	

Question **2**

Correct

Mark 1.00 out of 1.00

Compute the minimal and maximal Hamming distance between any two bitstrings in the set

$[111110, 110111, 110100, 100101]$.

$l_{\min} =$

Your last answer was interpreted as follows:

2

$l_{\max} =$

Your last answer was interpreted as follows:

4

Question **3**

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function given by $E(w_1 w_2 w_3) = w_1 w_2 w_3 w_4 w_5$, where

$$\begin{cases} w_4 = w_3 \\ w_5 = w_2 + w_1 \end{cases}$$

a) Find the generator matrix G for the encoding E

$G =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

b) Find the parity check matrix H for the encoding E .

$H =$

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

c) Let $r = 11001$ be a received message. Compute the syndrome of r with respect to the encoding E .

Syndrome of $r =$

Your last answer was interpreted as follows:

01

Syntax guide: Give your answers to **a)** and **b)** as matrices. Use a space character to separate entries in the same row, and line break to separate rows.

Syntax guide: Give your answer to **c)** as a bitstring. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 488

Signing code:

Your last answer was interpreted as follows:

39423

[◀ Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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